Servo Characterization for the SOFIA Modular Kinetic Surface

Marco Heimdal Castro Magaña, Alejandro Amador Franco, Karen Alejandra Guevara Vasquez, María Alicia Vela Contreras, Juan Pablo Contreras Quiñones, Jorge A. Lizarraga, Luis E. González-Jiménez, Luis F. Luque-Vega

***Abstract*—The SOFIA modular kinetic surface consists of 16 electromechanical modules arranged in a honeycomb configuration, each actuated by two servomotors, requiring the coordinated control of 32 actuators. Differences in the dynamic response of these servomotors may affect the synchronization and positioning performance of the system. This work proposes an experimental methodology for characterizing the dynamic response of the two servomotors in a single SOFIA module, temporarily removed from the complete structure for testing. A removable crown-shaped measurement device incorporating limit switches is proposed to establish repeatable angular reference positions and record the time intervals associated with servo movement. The control architecture consists of an ESP32-C3 microcontroller and two PCA9685 PWM drivers, providing independent control of the 32 actuators in the complete system. The proposed methodology includes repeatedmovement tests under defined operating conditions, analysis of the measured response times, and evaluation of an appropriate dynamic model for estimating effective servomotor time constants. The experimental scope is limited to the two servomotors of the selected module, with the aim of establishing a measurement procedure for evaluating their dynamic behavior.**

***Index Terms*—Servomotor characterization, dynamic response, time constant estimation, limit switches, modular robotics, kinetic surfaces.**

I. Introduction

SOFIA is a modular robotic installation composed of 16 mechatronic units arranged in a two-dimensional honeycomb configuration, with each unit providing two rotational degrees of freedom actuated by servomotors. The system aims to achieve coordinated control of the multiple actuators while enabling interaction through static and dynamic human gestures. For this purpose, SOFIA integrates embedded control, inter-module communication, and an external computer vision system for gesture recognition.

The SOFIA system requires the integration of mechanical actuation, servo control, embedded processing, and communication subsystems. Since each of the 16 units uses two servomotors, the complete demonstrator requires the coordinated control of 32 actuators and the generation of multiple PWM signals. The embedded controller must coordinate the desired actuator positions, while the communication architecture should support the future integration of additional SOFIA modules without significantly increasing wiring and control complexity. This work proposes an experimental method to estimate individual servo time constraints with a mechanical solution.

The authors are affiliated with the Instituto Tecnológico de Estudios Superiores de Occidente (ITESO), Tlaquepaque, Jalisco, Mexico.

A. Problem Statement and Research Question

Standard low-cost micro-servos exhibit manufacturing tolerances that result in non-uniform actuation speeds and transient response times across different units. In a 16-unit, 32-actuator modular kinetic surface like SOFIA, driving all servos with uniform control signals creates dynamic motion desynchronization and mechanical stress during normal use.

To address this limitation, this work investigates the following research question: *Can experimental estimation of individual servo mechanical time constants (τ_m) and their integration into a trajectory controller improve dynamic positional synchronization across a 32-actuator kinetic surface?*

This encompasses three primary objectives: (1) designing a PCB integrating an ESP32-C3 microcontroller, dual PCA9685 drivers, and RS-485 interfaces; (2) developing an automated micro-test protocol to capture step-response and extract individual time constants; and (3) integrating individual compensation parameters into the surface tracking controller to equalize dynamic settling times across all actuators.

B. Related shape-changing interface systems

Motorized arrays have been widely explored as a mechanism for generating dynamic physical surfaces. Relief introduced an actuated tabletop composed of 120 individually addressable motorized pins and emphasized a scalable hardware and software architecture [1]. This concept was extended through inFORM, where shape actuation was employed to generate dynamic physical affordances and constraints and to manipulate physical objects [2]. At a larger scale, TRANSFORM explored the transformation of static physical structures into dynamic systems driven by streams of data and energy [3]. These approaches demonstrate the potential of coordinated arrays of actuators, although their architectures primarily focus on large pin-based or installation-scale physical surfaces.

Alternative approaches have explored different actuation principles and modular configurations. PneUI demonstrated shape-changing interfaces based on pneumatically actuated soft composite materials, integrating sensing and active shape output within deformable structures [4]. In contrast, LineFORM investigated electromechanical shape transformation through serial chains of one-degree-of-freedom servo actuators [5]. This concept evolved toward greater hardware modularity in ChainFORM, where identical modules incorporate motor actuation, sensing, visual output, and embedded processing [6]. These systems highlight modularity and distributed actuation as relevant design characteristics for reconfigurable physical interfaces.

TABLE I
COMPARATIVE MATRIX OF RELATED SHAPE-CHANGING INTERFACE SYSTEMS

Reference	Method / technology	Metric / focus	Main result	Gap with respect to SOFIA
Relief [1]	Array of 120 motorized pins, scalable low-cost control	Actuation density, hardware/software scalability	Continuous surface capable of rendering and animating 3D shapes, with per-pin input sensing	Translational actuation only (1 DOF per pin); no independent rotation per unit
inFORM [2]	Pin-based shape display on a pre-existing platform	Dynamic physical affordances and constraints, object manipulation	Real-time direct-touch interaction and physical rendering of content	Same translational limitation as Relief; centralized architecture, not modular per unit
TRANSFORM [3]	Three large-scale pin displays, kinetic energy as input	Aesthetic transformation from static furniture to dynamic system	Over one thousand pins coordinated in real-time wave motion	Installation-scale system; no per-module orientable control or distributed communication between units
PneUI [4]	Pneumatically actuated multi-layer soft composite materials	Integrated input sensing and shape output in the material itself	Four functional applications (phicons, mobile, case, lamp) with controllable shape change	Soft/pneumatic actuation, not discrete rotational motion; difficult to scale to precise angular control
LineFORM [5]	Serial chain of one-DOF servo actuators	Line/curve-based interaction affordances	Cord, mobile, and body-constraint interfaces through line-shaped actuation	Serial (chain) configuration, not a two-dimensional array; 1 DOF per unit, not 2
ChainFORM [6]	Identical modules with servo, touch sensing, visual output and embedded processing	Hardware modularity and reconfigurability	Modular, reconfigurable platform with rich per-module I/O	Modularity demonstrated, but still a linear chain, not a 2D array with 2 DOF per node

TABLE II
BENCHMARKING AGAINST RELATED SHAPE-CHANGING SYSTEMS

Solution	Configuration	DOF/ unit	Modularity / scalability
inFORM [2]	Pin array on fixed platform	1 (translational)	Low – centralized control, not designed for modular expansion
LineFORM / ChainFORM [5], [6]	Serial chain of modules	1 per module	Medium – modular, but limited to a linear configuration
PneUI [4]	Pneumatic soft composite material	Continuous (non-discrete)	Low – difficult to discretize into individually addressable nodes
SOFIA (proposed)	2D array, honeycomb modules	**2 (rotational)**	**High – I2C bus + RS-485 designed for future module expansion**

The main technological characteristics and differences identified among the reviewed shape-changing interfaces are summarized in Table I.

In contrast to the reviewed shape-changing interfaces, SOFIA combines a two-dimensional honeycomb arrangement with two rotational degrees of freedom per unit. Unlike pin-based displays, which predominantly constrain motion to a single translational axis, or line-based modular systems organized as serial chains, each SOFIA unit can independently modify its orientation while remaining part of the same physical surface. This configuration introduces design challenges related to coordinated actuation, modularity, scalability, control architecture, electrical integration, and communication. Based on the reviewed literature, these characteristics were identified as relevant criteria for evaluating the technological architecture of the demonstrator. Table I summarizes the comparative analysis of the reviewed systems, while Table II contrasts SOFIA with the reviewed solutions using the established comparison criteria.

From the functional decomposition of the demonstrator and its comparison with the literature, several technological opportunities were identified. One is the development of a scalable control architecture capable of independently coordinating all actuators while reducing the processing load on the main controller. Another is the implementation of an inter-module communication architecture that allows multiple SOFIA modules to operate as part of a larger system. Additional opportunities include the implementation of software-defined motion limits and smoothing strategies to prevent abrupt servo movements and reduce mechanical stress.

Based on these opportunities, this work focuses on the proposed experimental characterization of servomotor dynamic response, supported by a preliminary electronic and control architecture for the SOFIA demonstrator. The selected hardware provides the PWM generation, embedded processing, and communication interfaces required by the test system, while the main research contribution is the methodology for measuring and analyzing individual actuator response. Automatic Modbus address assignment is outside the experimental scope.

II. MATERIALS AND METHODS

The SOFIA kinetic surface consists of 16 electromechanical modules arranged in a honeycomb configuration, each providing two rotational degrees of freedom actuated by MG90S servomotors [7], [8]. Although the complete system incorporates 32 actuators, the experimental characterization proposed in this study is limited to the two servomotors of a single module removed from the SOFIA structure.

The methodology is divided into three main stages: hardware configuration, experimental measurement, and dynamic response analysis. The first stage involves the selection and verification of the electronic components required for servomotor control, considering the established hardware requirements [9]. The second stage proposes the use of a removable crown-shaped measurement device equipped with limit switches to detect predefined angular positions during controlled movements. The third stage involves analyzing the recorded timing data and evaluating an appropriate dynamic model to estimate the effective time constant of each servomotor.

The experimental procedure is intended to provide repeatable measurements under defined operating conditions. Relevant variables include the commanded angular displacement, movement direction, supply voltage, and measured time intervals. These variables will be considered when evaluating differences in the dynamic response of the two servomotors.

TABLE III
SOFIA MAIN REQUIREMENTS

Requirement	Specification	Verification
Power	5 V servo supply	Compatible with MG90S range
Logic voltage	3.3 V MCU logic	Required for ESP32-C3 GPIO
Communication	WiFi/Bluetooth + I2C	Supported by ESP32-C3
PWM control	32 servo channels, 50 Hz	Two PCA9685 drivers required
Mechanical	16 modules, 2 DOF each	32 servos required

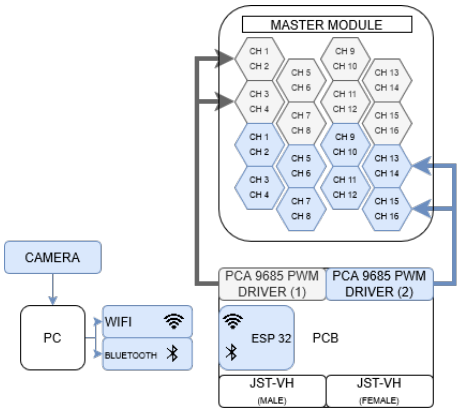


Fig. 1. Block diagram describing SOFIA components and connections.

The electronic architecture of the complete SOFIA system includes an ESP32-C3 microcontroller and two PCA9685 PWM drivers, which provide the 32 PWM channels required for actuator control [7], [10], [11]. However, only the two servomotors of the selected module will be evaluated experimentally in thisstudy.

The main electrical, communication, and mechanical requirements of the SOFIA demonstrator are summarized in Table III. These requirements are based on the system architecture and the established technical verification criteria [7], [9].

Figure 1 illustrates the system architecture and signal flow of the SOFIA demonstrator. Telemetry and motion commands are communicated wirelessly between a central host PC (equipped with an external vision camera) and the main PCB via Wi-Fi or Bluetooth protocols. Onboard processing is handled by an ESP32 microcontroller, which interfaces with two PCA9685 PWM drivers over I^2C to control a total of 32 servos. Driver 1 manages channels CH1–CH16 (grey group), while Driver 2 addresses channels CH17–CH32 (blue group, labeled CH1–CH16 on the second module) across the Master Module actuator array. Interconnectivity between boards is planned using JST-VH connectors.

A. ESP32-C3 Technical Analysis

The main technical specifications of the ESP32-C3 microcontroller are presented in Table IV.

B. PCA9685 Technical Analysis

The specifications of the PCA9685 PWM driver are summarized in Table V.

TABLE IV
ESP32-C3 MAIN SPECIFICATIONS AND DATASHEET VERIFICATION [10]

Parameter	**Value**	**Source (Doc / Ver / Pg)**	**Justification**	**Verification**
Supply Voltage	3.0–3.6 V	ESP32-C3 Datasheet v2.4, p. 18	Match system main power operating limits	Requires regulated 3.3 V supply
Wi-Fi	802.11 b/g/n	ESP32-C3 Datasheet v2.4, p. 2	Enable 2.4 GHz wireless transmission	Complies; integrated 2.4 GHz subsystem
Bluetooth	BLE 5	ESP32-C3 Datasheet v2.4, p. 2	Support low-energy configuration and monitoring	Complies; Bluetooth LE 5.0 supported
Logic Level	3.3 V GPIO	ESP32-C3 Datasheet v2.4, p. 19	Ensure digital signal levels meet peripheral logic thresholds	5 V logic cannot be applied directly without level shifting
Boot Pins	GPIO2, 8, 9	ESP32-C3 Datasheet v2.4, p. 12	Prevent unintended strapping state selection during power-up	PCB design hardware strapping verification required
Temperature	Up to 105 °C	ESP32-C3 Datasheet v2.4, p. 18	Ensure reliable continuous execution under thermal load	Suitable for industrial and laboratory operating environments

TABLE V
PCA9685 MAIN SPECIFICATIONS AND DATASHEET VERIFICATION [11]

Parameter	**Value**	**Source (Doc / Ver / Pg)**	**Justification**	**Verification**
Channels	16 PWM	PCA9685 Datasheet Rev. 7.5, p. 1	Control multiple servos with minimal MCU pin overhead	Two devices provide 32 outputs
Resolution	12 bits	PCA9685 Datasheet Rev. 7.5, p. 1	High-precision duty-cycle control for smooth motion	Complies; 4096-step resolution
Interface	I^2C FM+	PCA9685 Datasheet Rev. 7.5, p. 1	Standardized serial bus compatible with host controller	Compatible with ESP32-C3 I^2C peripheral
Supply Voltage	2.3–5.5 V	PCA9685 Datasheet Rev. 7.5, p. 52	Operate on standard 3.3 V logic power rail	Uses 3.3 V logic power supply
PWM Frequency	24–1526 Hz	PCA9685 Datasheet Rev. 7.5, p. 1	Match standard RC servo control signal requirements	Configured to 50 Hz output frequency
Addressing	Up to 62 ICs	PCA9685 Datasheet Rev. 7.5, p. 12	Support unique addressing for cascading multiple ICs	Configured with distinct fixed addresses (0x40 and 0x41)

C. MG90S Servo Technical Analysis

The nominal operating characteristics of the MG90S servomotor are presented in Table VI.

D. Checklist Verification

The preliminary component compatibility assessment is summarized in Table VII.

TABLE VI
MG90S MAIN SPECIFICATIONS AND DATASHEET VERIFICATION [8]

Parameter	Value	Source (Doc / Ver / Pg)	Justification	Verification
Voltage	4.8–6.0 V	TowerPro MG90S documentation, p. 1	Match system 5 V main power	Compatible with 5 V
Torque	1.8 kgf·cm at 4.8 V; 2.2 kgf·cm at 6.0 V	TowerPro MG90S documentation, p. 1	Provide required joint actuation torque under load	1.88 kgf·cm available at 5.0 V; validation required
Speed	0.10 s/60° at 4.8 V; 0.08 s/60° at 6.0 V	TowerPro MG90S documentation, p. 1	Ensure joint angular velocity meets dynamic motion profile	Reference value; complies for target control loop speed
Mass	13.4 g	TowerPro MG90S documentation, p. 1	Minimize total end-effector and leg inertia	Must be considered in structural mass budget and CAD model
PWM Frequency	50 Hz	TowerPro MG90S documentation, p. 1	Standard RC servo control signal frame timing (20 ms period)	Compatible with PCA9685 PWM output driver
Stall Current	Not reported	TowerPro MG90S documentation, p. 1	Sizing power supply current limits for worst-case loading	Not specified in datasheet; physical current measurement required

TABLE VII
VERIFICATION SUMMARY

Component	Status	Comment
ESP32-C3	**Complies**	Requires 3.3 V regulation and boot verification
PCA9685	**Complies**	Requires unique I2C addresses and 50 Hz configuration
MG90S	**Partial**	Torque and current require validation

E. RS-485 Logic-Level Compatibility

To enable communication with an additional set of 16 SOFIA modules, the proposed architecture incorporates the MAX13487E TTL485-V2.0 transceiver module. The MAX13487E provides half-duplex RS-485 communication with automatic direction control, reducing the need for additional transmission-direction control signals [17].

According to the manufacturer's datasheet, the MAX13487E operates with a nominal 5 V supply, whereas the ESP32-C3 uses 3.3 V GPIO logic [10], [17]. A preliminary electrical test identified a receiver-output voltage of approximately 5.00 V at the TXD terminal of the TTL485-V2.0 module. This voltage exceeds the permitted input range of the ESP32-C3.

To address this incompatibility, a resistive voltage divider was incorporated into the PCB schematic between the TTL485-V2.0 TXD output and the ESP32-C3 UART receive input. The divider consists of a 10 kOhm series resistor and an 18 kOhm resistor connected to ground, producing approximately 3.21 V from a nominal 5 V input.

The ESP32-C3 UART transmit output remains directly connected to the TTL485-V2.0 RXD input, since the MAX13487E specifies a minimum logic-high input voltage of 2.0 V [17].

TABLE VIII
SOFIA DEMONSTRATOR BILL OF MATERIALS (BOM) SUMMARY

Item	Component	Qty / Unit	Qty / Sys	Technical Description	Manufacturer	Location
1	Sofia003-P001-03	1	16	Sofia Module Base	In-house (LabDST)	ITESO, MX
2	MG90S / SG90 Servo	2	32	Micro Servo Motor	Tower Pro	Hong Kong
3	Sofia003-E002-01	1	16	Lower Joint Subassembly	In-house (LabDST)	ITESO, MX
3.1/4.4	Pro3	2	32	Servo Horn	Tower Pro	Hong Kong
3.2	Sofia003-P004-02	1	16	Lower Joint Support Left	In-house (LabDST)	ITESO, MX
3.3	Sofia003-P004-03	1	16	Lower Joint Support Right	In-house (LabDST)	ITESO, MX
3.4/3.5/ 4.3/7/8	M2 Screws (various)	10	160	Servo Arm Screw ($M2 \times 10/12$ mm)	Walfront	Shenzhen, CN
3.6	Axis 1 (9 × 9 mm)	1	16	Joint Shaft (9 mm × 9 mm)	Mae	Local
3.7	Sofia003-P002-02	1	16	Lower Joint Link	In-house (LabDST)	ITESO, MX
4	Sofia003-E003-01	1	16	Upper Joint Subassembly	In-house (LabDST)	ITESO, MX
4.1	Sofia003-P005-01	1	16	Upper Joint Support Left	In-house (LabDST)	ITESO, MX
4.2	Sofia003-P005-02	1	16	Upper Joint Support Right	In-house (LabDST)	ITESO, MX
5	Sofia003-P008-01	1	16	Sofia Top Module Linkage	In-house (LabDST)	ITESO, MX
6	Sofia003-P009-01	1	16	Top Platform Support	In-house (LabDST)	ITESO, MX
9	Inser M3x5x4	1	16	M3 Threaded Insert (5 × 4 mm)	ZWMSSLL	Shenzhen, CN
10	Sofia003-P000-01	1	16	Sofia Hexagonal Top Platform	In-house (LabDST)	ITESO, MX
–	ESP32-C3 SuperMini	–	1	Central MCU (Wi-Fi/BLE telemetry)	Espressif	Shanghai, CN
–	PCA9685 Servo Driver	–	2	16-channel 12-bit I^2C PWM driver board	NXP Semi	Eindhoven, NL
–	MAX13487E TTL485	–	1	Controller-side half-duplex RS-485 transceiver	Analog Devices	Wilmington, US
–	LED RGB	–	1	System state status indicator LED	Mexbit	Local
–	JST-VH Male	–	1	High-current power PCB header	JTAREA	Local
–	JST-VH Female	–	1	High-current power harness plug	JTAREA	Local

TABLE IX
TRADE-OFF ANALYSIS FOR ACTUATION

Alternative	Torque	Mass	Key Limitation / Feature	Sel.
MG996R [12]	High	55 g	Excessive size and current draw	
SG90 [13]	Low	9 g	Fragile nylon gears under load	
MG90S [8]*	Medium	13.4 g	Metal gears; optimal torque/size	✓

TABLE X
TRADE-OFF ANALYSIS FOR PWM EXPANSION

Alternative	Channels	Key Limitation / Feature	Sel.
Software PWM [10]	Limited	High MCU timer overhead; Limited GPIO	
Auxiliary MCU [14]	Flexible	Requires custom firmware and PCB space	
PCA9685 [11]*	16	Hardware 12-bit PWM over I^2C bus	✓

The voltage-divider circuit is included in the revised PCB schematic. Experimental verification of the assembled PCB remains pending.

F. Bill of Materials

The components required for the SOFIA demonstrator, including mechanical parts, actuators, and electronic devices, are summarized in Table VIII.

G. Justification of Technological Selection

The technological architecture was selected according to the functional and scalability requirements of the SOFIA demonstrator. MG90S servomotors as stated in Table IX were selected to provide controlled angular positioning for each

TABLE XI
TRADE-OFF ANALYSIS FOR MAIN CONTROLLER

Alternative	Telemetry	Key Limitation / Feature	Sel.
Arduino Nano [15]	None	8-bit AVR; lacks integrated wireless	
STM32F4 Core [14]	External	High compute; larger board area	
ESP32-C3 SuperMini [10]*	Native	32-bit RISC-V with Wi-Fi/BLE	✓

TABLE XII
TRADE-OFF ANALYSIS FOR INTER-MODULE COMMUNICATION

Alternative	Topology	Key Limitation / Feature	Sel.
Direct UART [10]	Point-to-Point	Poor multi-drop scalability	
CAN Bus [16]	Multi-drop	High protocol/driver complexity	
RS-485 (MAX13487E) [17]*	Multi-drop	Differential noise immunity + auto direction	✓

degree of freedom, while two PCA9685 drivers as stated in Table X were selected to provide the 32 PWM channels required by the system through a shared I2C bus. The ESP32-C3 SuperMini as stated in Table XI was selected as the main controller due to its integrated wireless connectivity and support for the required communication interfaces. Finally, RS-485 as stated in Table XII was selected for inter-module communication because its differential and multi-node architecture supports the future connection of multiple SOFIA modules.

H. Servo Mechanical Time Constant Estimation

The first-order dynamic model for the angular position response $\theta(t)$ to a step input is defined by:

$$\theta(t) = \theta_f + (\theta_0 - \theta_f)e^{-t/\tau_m} \tag{1}$$

where θ_0 represents the initial angular position, θ_f denotes the commanded step target angle, and τ_m is the servo mechanical time constant.

To characterize τ_m, no-load step responses of $60°$ will be applied while measuring the transient velocity profile. As noted in Table VI, the manufacturer datasheet [8] lists nominal angular speed (0.10 s/$60°$ at 4.8 V and 0.08 s/$60°$ at 6.0 V) but omits the explicit value for τ_m, necessitating experimental identification.

III. DESIGN AND IMPLEMENTATION OF THE TECHNOLOGICAL SOLUTION

Figure 2 shows the simulated response of the sixteen-module SOFIA structure while tracking the moving point. The sequence illustrates each module reorientation at different instants, where each module adjusts its two rotational degrees of freedom to maintain alignment with reference position.

Algorithm 1 summarizes the inverse kinematics and smoothing strategy used by each SOFIA module to orient its platform toward the moving target.

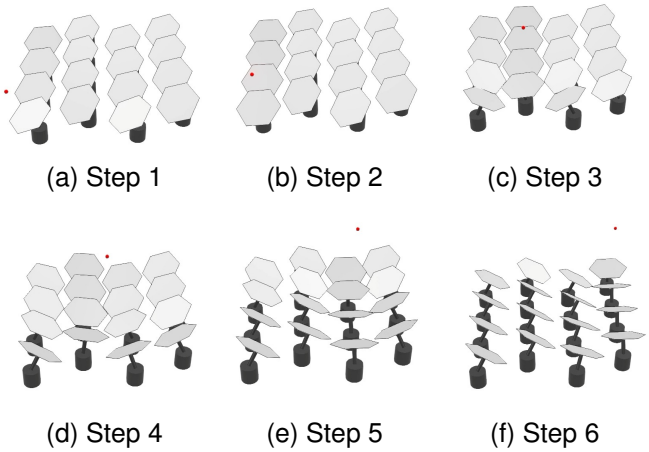


Fig. 2. Simulation of the SOFIA module tracking a moving reference point using inverse kinematics.

Algorithm 1 SOFIA module target tracking control

Require: Target position $\mathbf{p}$, module base position $\mathbf{r}_{0,2}$, joint limits $[q_{1\,\min}, q_{1\,\max}]$ and $[q_{2\,\min}, q_{2\,\max}]$, EMA smoothing factor $\alpha \in (0, 1]$

Ensure: Servo command angles q_1, q_2

$\mathbf{v} \leftarrow \mathbf{p} - \mathbf{r}_{0,2}$
$\hat{\mathbf{r}}_{2,p} \leftarrow \frac{\mathbf{v}}{\|\mathbf{v}\|} = [u_x, u_y, u_z]^\top$
$q_1^* \leftarrow \mathrm{atan2}(-u_y, u_z)$
$q_2^* \leftarrow \arcsin(u_x)$
$q_1^* \leftarrow \textsc{ClampAngle}(q_1^*, q_{1\,\min}, q_{1\,\max})$
$q_2^* \leftarrow \textsc{ClampAngle}(q_2^*, q_{2\,\min}, q_{2\,\max})$
$q_1 \leftarrow q_{1,\text{old}} + \alpha(q_1^* - q_{1,\text{old}})$ ▷ First-order EMA low-pass filter update
$q_2 \leftarrow q_{2,\text{old}} + \alpha(q_2^* - q_{2,\text{old}})$
Apply q_1, q_2 to servo motors
return q_1, q_2

IV. RESULTS

A. Experimental Setup

B. Servomotor Dynamic Response

C. Time Constant Estimation

V. DISCUSSION

VI. CONCLUSIONS

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